VIVEKANADA COLLEGE OF ENIGEERING & TECHNOLOGY
[A Unit of Vivekananda Vidyavardhaka Sangha, Putturᴼᴿ]
Affiliated to Visvesvararaya Technological University, Belagavi
Approved by AICTE, New Delhi & Govt of Karanataka

MINI PROJECT

SKILL ENCHANCEMENT COURSE(MMC258x)

WEATHER PREDICTION SYSTEM

Submitted To,

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INTRODUCTION

TITLE OF THE PROJECT – Weather Forecasting

1. INTRODUCTION OF THE PROJECT

Weather plays an important role in our daily life, influencing activities such as travel, agriculture, and outdoor planning. Accurate weather prediction helps people make informed decisions and avoid risks caused by sudden weather changes.

With the advancement of Machine Learning (ML), it is possible to analyze historical weather data and predict future weather conditions such as temperature, rainfall, and humidity. This project focuses on developing a predictive system that forecasts weather conditions using past data and ML algorithms.

2. PROBLEM STATEMENT

Weather conditions are often unpredictable, and sudden changes can affect daily activities and safety. Many people do not have access to simple tools that can provide accurate weather predictions based on data analysis.

This project aims to develop a machine learning-based system that predicts future weather conditions using historical weather data.

3. OBJECTIVES

- To develop a system that predicts weather conditions
- To analyze historical weather data such as temperature and humidity
- To apply machine learning algorithms for prediction
- To provide user-friendly output (e.g., predicted temperature or rainfall)

4. TOOLS AND TECHNOLOGY USED

- **Programming Language:** Python
- • **IDE:** VS Code
- **Libraries:**
    - Pandas
    - NumPy
    - Scikit-learn
    - matplotlib

5. CONCLUSION

This project demonstrates how machine learning can be used to predict weather conditions using historical data.The system helps users understand future weather trends,allowing better planning and decision-making.it shows the practical application of ML in solvind real-world problem related to weather forcasting.